

Universidad del Valle de Guatemala

Facultad de Ingeniería

Redes - Sección 10 - Semestre 2, 2026



Laboratorio 3 - Algoritmos de Enrutamiento

Planificación IP, VLSM, VLANs y enrutamiento

OSPF / EIGRP con redistribución

Integrante:

Carlos Daniel Estrada Vega - 20853

Guatemala, 6 de octubre de 2026

Laboratorio 3 - Evidencias de enrutamiento

Packet Tracer 9.0 | Evidencias reales de configuración y conectividad | 5 de octubre de 2026

Archivo principal: Lab3_Enrutamiento.pkt. Cinco routers, cuatro switches y seis PCs de prueba. OSPF área 0 y EIGRP AS 100.

Limitación pendiente: el IOS simulado rechazó summary-address de OSPF, route-map y distribute-list. En el archivo principal, CENTRAL publica 172.16.0.0/24 mediante una ruta estática Null0 con distancia 250, redistribuida como OSPF E1. El tráfico utiliza las rutas EIGRP más específicas. Esto logra conectividad y un resumen, pero NO cumple literalmente la redistribución dinámica EIGRP hacia OSPF solicitada.

Lab3_Redistribucion_Directa.pkt conserva redistribución dinámica mutua, pero anuncia las LAN EIGRP específicas hacia OSPF y presenta realimentación del agregado OSPF. Las dos variantes tienen esta diferencia documentada; ninguna se presenta como cumplimiento total del requisito avanzado.

OSPF: enlace directo costo 20 y vía CENTRAL costo 10 + 10. Las rutas a LAN remota cuestan 21 en ambos caminos, sumando costo LAN 1.

EIGRP: variance 4. De EIGRP-1 a EIGRP-2: métricas 102656 y 310016 (3.020:1); de EIGRP-2 a EIGRP-1: 104960 y 312320 (2.976:1). Ambas rutas alternativas cumplen distancia reportada menor que distancia factible. Es evidencia de métricas/configuración, no una medición de tráfico 75/25.

Las evidencias EIGRP se obtuvieron antes del cambio de origen del resumen OSPF; las métricas, vecinos y rutas internas EIGRP no cambiaron. Las tablas OSPF, CENTRAL, ping y tracert corresponden a la variante principal final. El PDF1 debe ser escrito a mano por los estudiantes.

R-OSPF-1 - tabla de enrutamiento

Salida real: show ip route

```
R-OSPF-1#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.1/32 is directly connected, Serial0/1/0
C       10.0.0.4/30 is directly connected, Serial0/1/1
L       10.0.0.6/32 is directly connected, Serial0/1/1
O       10.0.0.8/30 [110/20] via 10.0.0.5, 00:24:02, Serial0/1/1
    172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.5, 00:04:10, Serial0/1/1
O       192.168.0.0/24 [110/21] via 10.0.0.2, 00:23:47, Serial0/1/0
        [110/21] via 10.0.0.5, 00:23:47, Serial0/1/1
    192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

```
R-OSPF-1#
```

R-OSPF-2 - tabla de enrutamiento

Salida real: show ip route

R-OSPF-2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/1/0
L       10.0.0.2/32 is directly connected, Serial0/1/0
O       10.0.0.4/30 [110/20] via 10.0.0.9, 00:21:35, Serial0/1/1
C       10.0.0.8/30 is directly connected, Serial0/1/1
L       10.0.0.10/32 is directly connected, Serial0/1/1
    172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/30] via 10.0.0.9, 00:01:54, Serial0/1/1
    192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/21] via 10.0.0.1, 00:21:35, Serial0/1/0
        [110/21] via 10.0.0.9, 00:21:35, Serial0/1/1
O       192.168.1.64/28 [110/21] via 10.0.0.1, 00:21:35, Serial0/1/0
        [110/21] via 10.0.0.9, 00:21:35, Serial0/1/1
```

R-OSPF-2#

R-CENTRAL - tabla de enrutamiento

Salida real: show ip route

```
R-CENTRAL#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
    10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/30] via 10.0.0.6, 00:19:38, Serial0/0/0
        [110/30] via 10.0.0.10, 00:19:38, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.5/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/104960] via 10.0.0.18, 00:18:10, Serial0/1/0
        [90/104960] via 10.0.0.22, 00:18:04, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.17/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.21/32 is directly connected, Serial0/1/1
    172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/28416] via 10.0.0.22, 00:18:05, Serial0/1/1
D       172.16.0.128/26 [90/30720] via 10.0.0.18, 00:18:49, Serial0/1/0
D       172.16.0.192/26 [90/30720] via 10.0.0.18, 00:18:49, Serial0/1/0
D       192.168.0.0/23 is a summary, 00:18:59, Null0
O       192.168.0.0/24 [110/11] via 10.0.0.10, 00:19:38, Serial0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/11] via 10.0.0.6, 00:20:48, Serial0/0/0
O       192.168.1.64/28 [110/11] via 10.0.0.6, 00:20:48, Serial0/0/0
```

```
R-CENTRAL#
```

R-EIGRP-1 - tabla de enrutamiento

Salida real: show ip route

```
R-EIGRP-1#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX   10.0.0.0/30 [170/332800] via 10.0.0.17, 00:12:05, Serial0/1/1
D EX   10.0.0.4/30 [170/332800] via 10.0.0.17, 00:12:05, Serial0/1/1
D EX   10.0.0.8/30 [170/332800] via 10.0.0.17, 00:12:05, Serial0/1/1
C       10.0.0.12/30 is directly connected, Serial0/1/0
L       10.0.0.13/32 is directly connected, Serial0/1/0
C       10.0.0.16/30 is directly connected, Serial0/1/1
L       10.0.0.18/32 is directly connected, Serial0/1/1
D       10.0.0.20/30 [90/309760] via 10.0.0.17, 00:11:28, Serial0/1/1
          [90/384000] via 10.0.0.14, 00:11:22, Serial0/1/0
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
D       172.16.0.0/25 [90/102656] via 10.0.0.14, 00:11:22, Serial0/1/0
          [90/310016] via 10.0.0.17, 00:11:22, Serial0/1/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/26 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D       192.168.0.0/23 [90/819200] via 10.0.0.17, 00:12:05, Serial0/1/1
```

```
R-EIGRP-1#
```

R-EIGRP-2 - tabla de enrutamiento

Salida real: show ip route

R-EIGRP-2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX   10.0.0.0/30 [170/332800] via 10.0.0.21, 00:13:21, Serial0/1/1
D EX   10.0.0.4/30 [170/332800] via 10.0.0.21, 00:13:21, Serial0/1/1
D EX   10.0.0.8/30 [170/332800] via 10.0.0.21, 00:13:21, Serial0/1/1
C       10.0.0.12/30 is directly connected, Serial0/1/0
L       10.0.0.14/32 is directly connected, Serial0/1/0
D       10.0.0.16/30 [90/309760] via 10.0.0.21, 00:13:21, Serial0/1/1
        [90/384000] via 10.0.0.13, 00:13:22, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
    172.16.0.0/16 is variably subnetted, 4 subnets, 3 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/104960] via 10.0.0.13, 00:13:22, Serial0/1/0
        [90/312320] via 10.0.0.21, 00:13:21, Serial0/1/1
D       172.16.0.192/26 [90/104960] via 10.0.0.13, 00:13:22, Serial0/1/0
        [90/312320] via 10.0.0.21, 00:13:21, Serial0/1/1
D      192.168.0.0/23 [90/819200] via 10.0.0.21, 00:13:21, Serial0/1/1
```

R-EIGRP-2#

R-OSPF-1 - interfaces OSPF

Las interfaces seriales muestran costos 20 y 10. El segundo tramo vía CENTRAL añade 10; las dos rutas se verifican en la tabla anterior.

```
R-OSPF-1#show ip ospf interface
```

```
GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.65
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Internet address is 10.0.0.6/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
R-OSPF-1#
```

R-OSPF-2 - interfaces OSPF

Las interfaces seriales muestran costos 20 y 10. El segundo tramo vía CENTRAL añade 10; las dos rutas se verifican en la tabla anterior.

```
R-OSPF-2#show ip ospf interface
```

```
GigabitEthernet0/0 is up, line protocol is up
  Internet address is 192.168.0.1/24, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 2.2.2.2, Interface address 192.168.0.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/1/0 is up, line protocol is up
  Internet address is 10.0.0.2/30, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type POINT-TO-POINT, Cost: 20
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 1.1.1.1
  Suppress hello for 0 neighbor(s)
Serial0/1/1 is up, line protocol is up
  Internet address is 10.0.0.10/30, Area 0
  Process ID 1, Router ID 2.2.2.2, Network Type POINT-TO-POINT, Cost: 10
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
R-OSPF-2#
```

R-EIGRP-1 - topologia

R-EIGRP-1#show ip eigrp topology

IP-EIGRP Topology Table for AS 100/ID(172.16.0.193)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - Reply status

```
P 10.0.0.0/30, 1 successors, FD is 332800
    via Rstatic (332800/51200)
P 10.0.0.4/30, 1 successors, FD is 332800
    via Rstatic (332800/51200)
P 10.0.0.8/30, 1 successors, FD is 332800
    via Rstatic (332800/51200)
P 10.0.0.12/30, 1 successors, FD is 102400
    via Connected, Serial0/1/0
P 10.0.0.16/30, 1 successors, FD is 307200
    via Connected, Serial0/1/1
P 10.0.0.20/30, 2 successors, FD is 309760
    via 10.0.0.17 (309760/28160), Serial0/1/1
    via 10.0.0.14 (384000/307200), Serial0/1/0
P 172.16.0.0/25, 2 successors, FD is 102656
    via 10.0.0.14 (102656/2816), Serial0/1/0
    via 10.0.0.17 (310016/28416), Serial0/1/1
P 172.16.0.128/26, 1 successors, FD is 28160
    via Connected, GigabitEthernet0/0.10
P 172.16.0.192/26, 1 successors, FD is 28160
    via Connected, GigabitEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 819200
    via 10.0.0.17 (819200/537600), Serial0/1/1
R-EIGRP-1#
```

R-EIGRP-1 - protocolos

R-EIGRP-1#show ip protocols

```
Routing Protocol is "eigrp 100 "  
  Outgoing update filter list for all interfaces is not set  
  Incoming update filter list for all interfaces is not set  
  Default networks flagged in outgoing updates  
  Default networks accepted from incoming updates  
  Redistributing: eigrp 100  
  EIGRP-IPv4 Protocol for AS(100)  
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0  
    NSF-aware route hold timer is 240  
    Router-ID: 10.0.0.13  
    Topology : 0 (base)  
      Active Timer: 3 min  
      Distance: internal 90 external 170  
      Maximum path: 4  
      Maximum hopcount 100  
      Maximum metric variance 4  
  
Automatic Summarization: disabled  
Automatic address summarization:  
Maximum path: 4  
Routing for Networks:  
  172.16.0.0/20  
  10.0.0.13/32  
  10.0.0.18/32  
Passive Interface(s):  
  Vlan1  
  GigabitEthernet0/0  
  GigabitEthernet0/1  
  GigabitEthernet0/0.10  
  GigabitEthernet0/0.20  
Routing Information Sources:  
  Gateway         Distance        Last Update  
  10.0.0.17        90              3157625  
  10.0.0.14        90              3202564  
Distance: internal 90 external 170
```

R-EIGRP-1#

R-EIGRP-2 - topologia

R-EIGRP-2#show ip eigrp topology

IP-EIGRP Topology Table for AS 100/ID(172.16.0.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - Reply status

```
P 10.0.0.0/30, 1 successors, FD is 332800
  via Rstatic (332800/51200)
P 10.0.0.4/30, 1 successors, FD is 332800
  via Rstatic (332800/51200)
P 10.0.0.8/30, 1 successors, FD is 332800
  via Rstatic (332800/51200)
P 10.0.0.12/30, 1 successors, FD is 102400
  via Connected, Serial0/1/0
P 10.0.0.16/30, 2 successors, FD is 309760
  via 10.0.0.21 (309760/28160), Serial0/1/1
  via 10.0.0.13 (384000/307200), Serial0/1/0
P 10.0.0.20/30, 1 successors, FD is 307200
  via Connected, Serial0/1/1
P 172.16.0.0/25, 1 successors, FD is 2816
  via Connected, GigabitEthernet0/0
P 172.16.0.128/26, 2 successors, FD is 104960
  via 10.0.0.13 (104960/28160), Serial0/1/0
  via 10.0.0.21 (312320/30720), Serial0/1/1
P 172.16.0.192/26, 2 successors, FD is 104960
  via 10.0.0.13 (104960/28160), Serial0/1/0
  via 10.0.0.21 (312320/30720), Serial0/1/1
P 192.168.0.0/23, 1 successors, FD is 819200
  via 10.0.0.21 (819200/537600), Serial0/1/1
R-EIGRP-2#
```

R-EIGRP-2 - protocolos

R-EIGRP-2#show ip protocols

```
Routing Protocol is "eigrp 100 "  
  Outgoing update filter list for all interfaces is not set  
  Incoming update filter list for all interfaces is not set  
  Default networks flagged in outgoing updates  
  Default networks accepted from incoming updates  
  Redistributing: eigrp 100  
  EIGRP-IPv4 Protocol for AS(100)  
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0  
    NSF-aware route hold timer is 240  
    Router-ID: 10.0.0.14  
    Topology : 0 (base)  
      Active Timer: 3 min  
      Distance: internal 90 external 170  
      Maximum path: 4  
      Maximum hopcount 100  
      Maximum metric variance 4  
  
  Automatic Summarization: disabled  
  Automatic address summarization:  
  Maximum path: 4  
  Routing for Networks:  
    172.16.0.0/20  
    10.0.0.14/32  
    10.0.0.22/32  
  Passive Interface(s):  
    Vlan1  
    GigabitEthernet0/0  
    GigabitEthernet0/1  
  Routing Information Sources:  
    Gateway         Distance      Last Update  
    10.0.0.21        90            3187346  
    10.0.0.13        90            3187953  
  Distance: internal 90 external 170
```

R-EIGRP-2#

Conectividad extremo a extremo

Ping desde PC0 (192.168.1.2) a PC5 (172.16.0.2): cuatro respuestas y 0% de pérdida.

Cisco Packet Tracer PC Command Line 1.0

```
C:\>ping 172.16.0.2
```

Pinging 172.16.0.2 with 32 bytes of data:

```
Reply from 172.16.0.2: bytes=32 time<1ms TTL=125
```

```
Reply from 172.16.0.2: bytes=32 time<1ms TTL=125
```

```
Reply from 172.16.0.2: bytes=32 time<1ms TTL=125
```

```
Reply from 172.16.0.2: bytes=32 time<1ms TTL=125
```

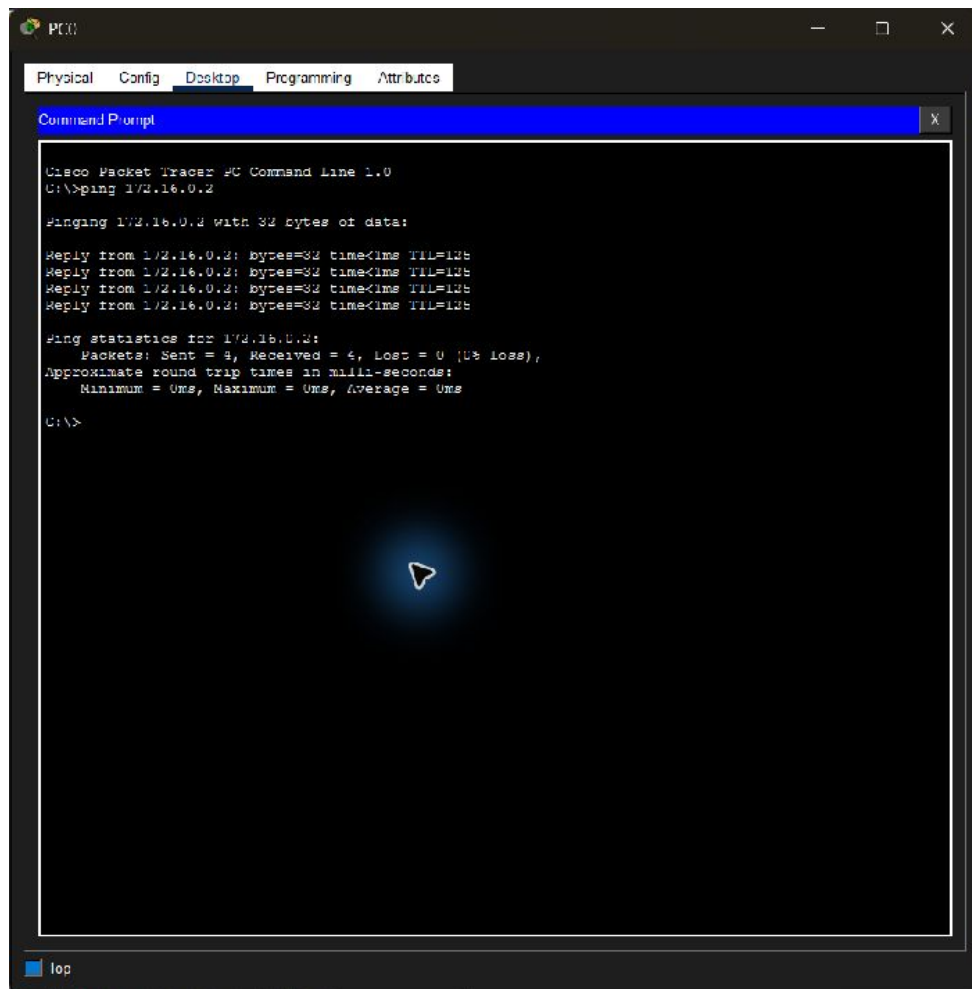
Ping statistics for 172.16.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

```
C:\>
```



Conectividad extremo a extremo

Tracert desde PC0 a PC5: recorrido por la frontera CENTRAL hasta el extremo EIGRP.

Cisco Packet Tracer PC Command Line 1.0

```
C:\>ping 172.16.0.2
```

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time<1ms TTL=125

Reply from 172.16.0.2: bytes=32 time<1ms TTL=125

Reply from 172.16.0.2: bytes=32 time<1ms TTL=125

Reply from 172.16.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 172.16.0.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

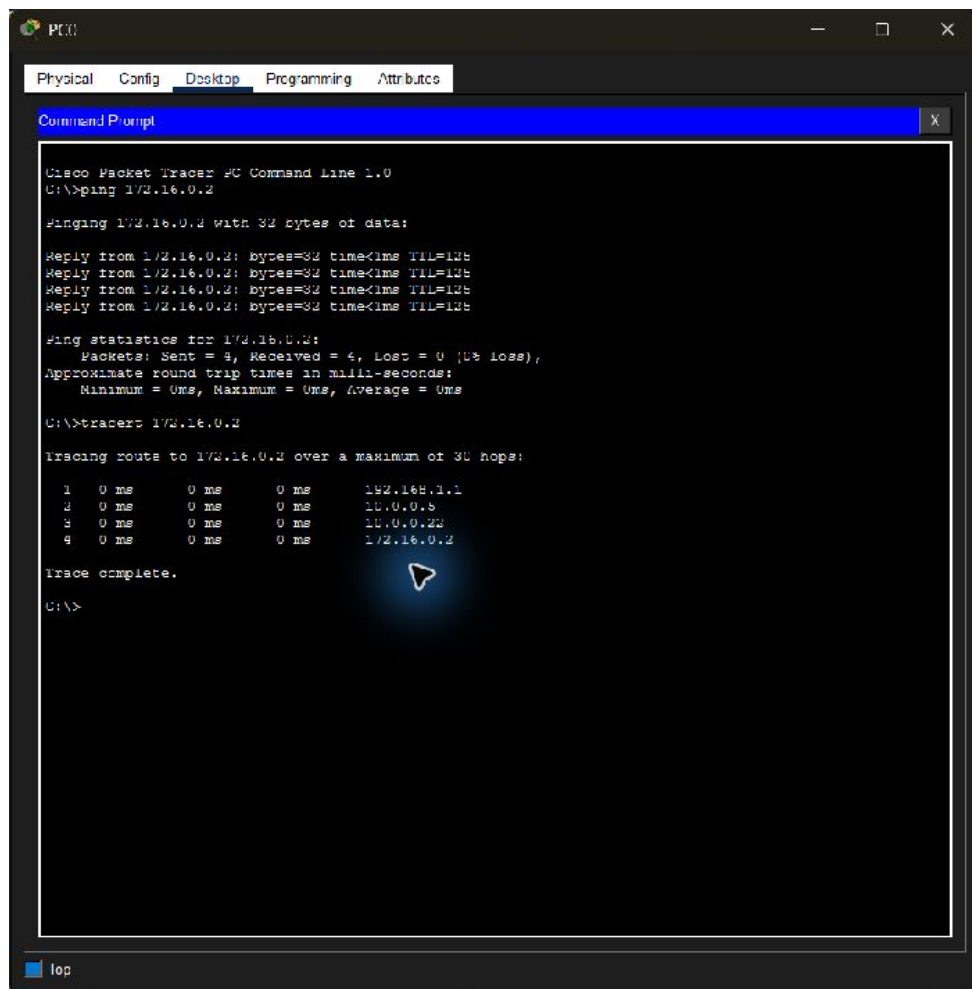
```
C:\>tracert 172.16.0.2
```

Tracing route to 172.16.0.2 over a maximum of 30 hops:

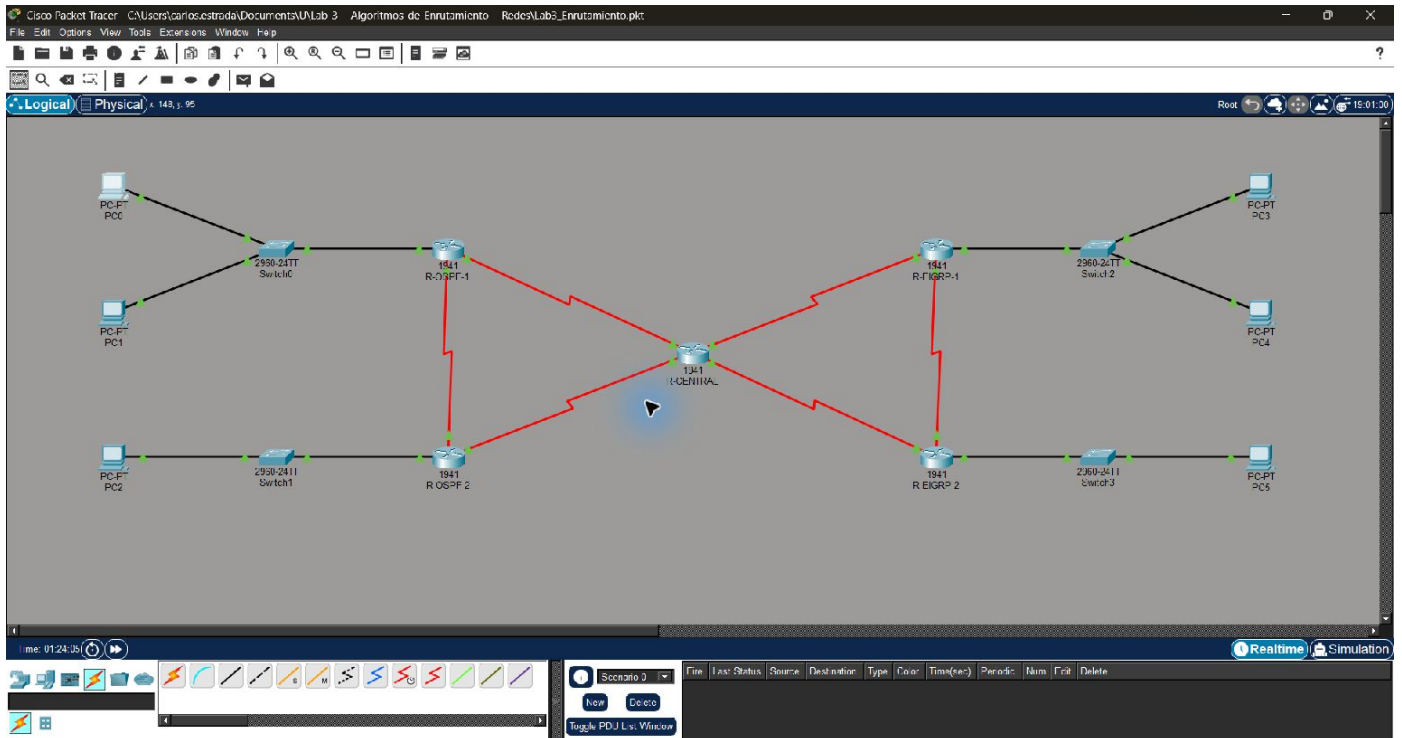
1	0 ms	0 ms	0 ms	192.168.1.1
2	0 ms	0 ms	0 ms	10.0.0.5
3	0 ms	0 ms	0 ms	10.0.0.22
4	0 ms	0 ms	0 ms	172.16.0.2

Trace complete.

```
C:\>
```



Topología construida



Los seis enlaces seriales y las conexiones LAN están activos. PC0 y PC1 prueban las VLAN 10 y 30 de OSPF-1; PC3 y PC4 prueban las VLAN 10 y 20 de EIGRP-1.

La página 2 del enunciado menciona tres VLAN en EIGRP-1, pero la tabla detallada de página 3 define únicamente dos. Se implementaron las dos especificadas en esa tabla; conviene aclarar la discrepancia con el docente.